# How Many Documents Cover a Corpus

**Anonymous ACL submission**

## Abstract

A sample drawn from a survey population comes with a margin of error and a formula for the sample size a target precision needs. A sample drawn from a corpus to annotate or to summarize comes with neither. We supply both for one question: what share of the corpus lies within a chosen cosine distance $r$ of some sampled document, in a fixed embedding space. The share that does not is the missing mass at resolution $r$. Its estimator $G$ needs no labels: it is the share of sampled documents with no other sampled document within $r$ (Maurer, 2022). Under the classical nearest neighbour law, $G$ should depend on sample size $n$ and resolution $r$ only through $nr^d$, with $d$ the intrinsic dimension of the embedded corpus. We test this on seventeen pairs of corpus and encoder, including forum, census and political social media text. Where the law holds, a fit on 1,000 documents predicts $G$ at seven times that size with an out of sample $R^2$ above 0.92. The sample size the law prescribes for a target isolation rate falls within a factor of 1.5 of the empirical one in 77 of 83 cases. The ecological Chao and Jost extrapolation errs two to nine times more. For a finite corpus of $N$ documents the missing mass is $G\,(1 - n/N)$, to a mean error of 0.012 at sampling fractions up to 0.7. The law fails on corpora whose effective dimension changes with the radius, such as slogans beside discussion, and a fit inside the pilot flags them before the sample is drawn. Calibrated to mean the same specific content, the formula shows that a few hundred posts cover half of a repetitive corpus, while for forum questions and stories no feasible sample does.

## 1 Introduction

Large collections of text are sampled before they are studied. An annotation team codes a few thousand posts out of millions, a language model summarizes a few hundred documents because the corpus does not fit its context, and a topic model is fitted to a slice because the full run is slow. The sampler of a human population has a settled answer to the question such a slice raises. Given the population variance $\sigma^2$ and a tolerated error $e$, the formula $n = z^2\sigma^2/e^2$ gives the sample size, and the estimate comes with a margin of error (Cochran, 1977; Kish, 1965). The sampler of a corpus has no such formula. The sample size is set by budget, and whether the slice covers the corpus is left unasked.

The question has an exact form once a corpus is a set of points. Embed each document, choose a distance $r$ that says what counts as the same content, and ask what share of the corpus lies farther than $r$ from every sampled document. We call that share the missing mass at resolution $r$. In the discrete case, where two documents either share a category or do not, it is the quantity Good and Turing estimated from the share of draws whose category was seen once (Good, 1953). Ecologists built a sampling theory on that estimate: a rarefaction curve gives the coverage a smaller sample would have had, an extrapolation curve gives the coverage a larger sample will have, and together they say how many more units a target coverage needs (Chao and Jost, 2012). Maurer (2022) carried the estimator into metric spaces. His statistic $G$ is the share of sampled points with no other sampled point within $r$, and he proved bounds on its bias and variance. The estimator has had no application to text, and Maurer's treatment says nothing about how $G$ falls as the sample grows.

How $G$ falls follows from a classical result. For points drawn from a density on a $d$-dimensional set, the chance that a point has no neighbour within $r$ among $n$ others decays like $\exp(-c\,n\,r^d)$, the law behind nearest neighbour estimators of intrinsic dimension (Levina and Bickel, 2005; Facco et al., 2017) and behind the isolated vertices of a random geometric graph (Penrose, 2003). We take this law, which has not been used to plan samples, and test whether it holds well enough on embedded text to

set a sample size. Three results carry the paper.

1. On corpora where the law holds, $G$ depends on $n$ and $r$ through $nr^d$, with $d$ the intrinsic dimension estimated from a pilot of 2,000 documents and two shape constants fitted on the pilot. A fit on 1,000 documents predicts $G$ at seven times that size with an out of sample $R^2$ above 0.92. The sample size it prescribes for a target isolation rate falls within a factor of 1.5 of the empirical sample size in 77 of 83 cases, over ten such pairs of corpus and encoder. The Chao and Jost extrapolation, the standard ecological tool, errs two to nine times more within its own range.

2. For a finite corpus of $N$ documents the missing mass is $G\,(1-n/N)$, because the sampled documents are covered by definition. The correction cuts the mean error tenfold, to 0.012.

3. The law fails on some corpora of political posts, where the effective dimension changes with the radius, slogans beside discussion. A fit inside the pilot flags every one of them before the sample is drawn, and on them the formula's predictions fall within a factor of 1.5 in only 10 of 23 cases.

With the resolution calibrated to mean "the same specific content", the formula separates two kinds of corpus. On corpora that repeat a few claims, a few hundred documents cover half of the content. On forum questions, stories and Truth Social threads, the required sample exceeds the corpus by orders of magnitude, so no sample covers the specific content and only reading every document does.

In practice the procedure is short. A user embeds a pilot, picks the resolution $r$ that matches the question by judging a few hundred pairs, estimates $d$ and the two constants, checks the fit, and takes the sample size from the formula. After the sample is drawn, $G\,(1-n/N)$ reports the share of the corpus it leaves uncovered.

## 2 Background

### 2.1 Sample coverage in the discrete case

A sample of $n$ draws from a population of categories covers the categories it has seen. The mass it has missed is the total probability of the unseen categories. Let $f_k$ be the number of categories that appeared exactly $k$ times in the sample. Good (1953) showed that $f_1/n$, the share of draws whose category appeared once, estimates the missing mass with small bias, and Gale and Sampson (1995) made the estimator a standard tool for word frequencies. Good and Toulmin (1956) extended it to the coverage of an enlarged sample. Ecologists took the estimator as the basis of a sampling theory for species. Chao (1984) bounded the number of unseen species from $f_1$ and $f_2$, and Chao and Jost (2012) wrote the expected coverage of a sample enlarged by $m$ draws as a closed form in $n$, $f_1$ and $f_2$, so that a survey of a habitat can be stopped at a target coverage. Rarefaction and extrapolation of coverage became one framework in ecology (Colwell et al., 2012; Chao et al., 2014). Efron and Thisted (1976) asked how many words Shakespeare knew with the same machinery, and Orlitsky et al. (2016) gave the optimal prediction of the number of unseen species for a sample up to about $n \log n$ times larger than the one in hand.

The estimators have reached text and machine learning a few times. Hajič and Moss (2025) apply the Chao bound to a catalogue of chant to say how far a collection is from complete. Tran et al. (2017) fit species models to the answers to an open ended survey question to predict when new answers stop appearing. Chung et al. (2018) count the unknown examples of a training set with the same estimator, a species being a unique example, and Pal et al. (2026) estimate the mass of deployment states a classifier has never seen. In all four a species is a discrete label and two items either match or do not.

### 2.2 The missing mass in a metric space

Maurer (2022) removed the discreteness. In a metric space with distance $d(\cdot,\cdot)$, the missing mass of a sample $X_1,\dots,X_n$ at scale $r$ is the probability that a new draw lies farther than $r$ from every $X_i$. Maurer calls it the conditional missing mass because it is conditional on the sample drawn. His estimator is

$$G(X,r) = \frac{1}{n}\,\big|\{k : d(X_k, X_i) > r \text{ for all } i \neq k\}\big|, \tag{1}$$

the share of sample points with no other sample point within $r$. In the discrete case, with $d(x,y) = 1$ for $x \neq y$ and $r < 1$, it is $f_1/n$. Maurer proved that $G$ exceeds the conditional missing mass by at most $1/n$ in expectation and that the variance of $G$ minus the conditional missing mass is at most $3/n$ (his Theorem 2.1). The paper is theory. It sketches anomaly detection and coding

as applications, and it contains no data, no embeddings and no statement about how $G$ falls with $n$. That dependence is the subject of the nearest neighbour distance law: for a density $f$ on a set of dimension $d$, the probability that a point has no neighbour within $r$ among $n$ others is close to $\exp(-nf(x)\, v_d\, r^d)$, with $v_d$ the volume of the unit ball (Evans et al., 2002; Penrose, 2003). The same law underlies the correlation dimension of Grassberger and Procaccia (1983) and the estimators of Levina and Bickel (2005) and Facco et al. (2017). The nearest ecological construction is the attribute diversity of Chao et al. (2019), in which two species closer than a threshold $\tau$ count as partly the same and diversity is reported as a function of $\tau$. We treat $r$ the same way, as a resolution the user chooses.

### 2.3 Diversity of a sample in embedding space

Several measures describe how spread out a set of embedded points is. The Vendi score is the exponential of the entropy of the eigenvalues of the sample's cosine kernel matrix, an effective number of distinct items (Friedman and Dieng, 2023), and its cousins extend it to the orders of a Hill number (Pasarkar and Dieng, 2024). The precision and recall of Kynkäänniemi et al. (2019) count generated points inside nearest neighbour balls around a reference set, the density and coverage metrics of Naeem et al. (2020) make those counts robust to the reference set's own spread, and MAUVE compares two distributions on a quantized embedding space (Pillutla et al., 2021). Each describes the sample in hand or compares two samples. None asks what a larger sample would cover, and none has a parameter in the role of a population variance. Core-set selection chooses points to minimise the largest distance from any point to the chosen set (Sener and Savarese, 2018), and semantic deduplication removes points within a small distance of another (Abbas et al., 2023; Lee et al., 2022). Both need the full embedded corpus, and neither says how many points a target needs. The intrinsic dimension of learned representations has been measured for images (Pope et al., 2021), for the layers of deep networks (Ansuini et al., 2019) and for text, where it separates human from generated writing (Tulchinskii et al., 2023). We use it as the corpus parameter that sets a sample size.

### 2.4 Saturation in practice

Where a sample size for text is set at all, it is set by a stopping rule. Qualitative researchers stop interviewing when new interviews add no new code. Guest et al. (2006) found that point at twelve interviews, and a review of 23 empirical tests by Hennink and Kaiser (2022) puts it at 9 to 17. De Paoli and Mathis (2024) carry the rule to coding by a language model and track the slope of the count of unique codes. TopicGPT reports that its topic count levels off after about 600 documents and stops when 200 documents add no topic (Pham et al., 2024). Maeda et al. (2025) fit an LDA model with eight topics to 522 open ended survey responses and find that a sample of 100 responses recovers over 60% of the topics found on the full corpus and 250 recover about 70%. These are observations of a coverage curve. The present paper gives the curve a parameter, the intrinsic dimension $d$, and an estimator.

## 3 Method

### 3.1 Setting and estimand

A corpus is a finite set of $N$ documents $x_1, \ldots, x_N$. Each is embedded by a fixed encoder and normalised, and the distance between two documents is the cosine distance of their embeddings. A sample $S$ is a set of $n$ documents drawn without replacement. The user chooses a resolution $r$: two documents within $r$ of each other count as the same content for the purpose at hand. Contextual embeddings fill a narrow cone of the space (Ethayarajh, 2019), so the same cosine distance means different things for different encoders and corpora, and Section 5.6 calibrates $r$ against judgments of content. The quantity to estimate, for the sample actually drawn, is the missing mass at resolution $r$,

$$M(S, r) = \frac{1}{N} \left| \{ j : d(x_j, x_i) > r \text{ for all } i \in S \} \right|, \tag{2}$$

the share of corpus documents with no sampled document within $r$. Coverage is $1 - M$. The estimator is Maurer's $G$ of Equation 1, computed on the sample alone from each sampled document's distance to its nearest sampled neighbour. We write $G(n, r)$ for its value on a sample of size $n$.

The estimand is the finite corpus in hand. Sampling exists because the corpus is too large to annotate or to place in a model's context, and the full corpus is available to check the estimate.

### 3.2 The collapse law

Suppose the embedded documents lie near a set of intrinsic dimension $d$. The mass of a ball of radius

$r$ around document $x$ is then about $c(x)\, r^d$, where $c(x)$ is proportional to the local density, and the probability that none of $n$ sampled documents falls in the ball is about $\exp(-c(x)\, n\, r^d)$. Averaging over documents gives

$$\mathbb{E}\, G(n, r) \approx \mathbb{E}_x\big[\exp(-c(x)\, n\, r^d)\big] = \mathcal{L}_c(n\, r^d), \tag{3}$$

the Laplace transform of the distribution of $c$ evaluated at $nr^d$. Whatever the density, $G$ depends on $n$ and $r$ through the product $nr^d$ alone, so long as one $d$ holds across the radii of interest. The shape of the curve depends on how $c$ varies: a constant $c$ gives a pure exponential, and a spread of $c$ gives a curve that falls more slowly. We describe the shape by a stretched exponential, $\exp(-(\kappa\, n\, r^d)^b)$ with $b \le 1$, which is the Laplace transform of a particular mixing distribution, and fit

$$\log\big(-\log G(n, r)\big) = a + b \log\big(n\, r^d\big), \tag{4}$$

with $d$ estimated beforehand and $a$ and $b$ fitted by least squares. The grid has eleven sample sizes from 100 to 7,000 (capped at $0.7N$), twelve radii spaced geometrically from the 5th to the 99th percentile of a pilot's nearest neighbour distances, and five independent draws per sample size. Only cells with $0.02 < G < 0.98$ enter the fit. The law fails when the dimension itself changes with the radius, because no single $d$ then aligns the curves for different $r$.

The dimension $d$ comes from the TwoNN estimator of Facco et al. (2017), which uses the ratio of each point's second to first nearest neighbour distance, averaged over five pilots of 2,000 documents. As a check that owes nothing to TwoNN, we also regress $\log(-\log G)$ on $\log n$ and $\log r$ separately. If the law holds, the coefficient on $\log r$ is $b\, d$ and the coefficient on $\log n$ is $b$, so their ratio estimates $d$.

Solving Equation 4 for $n$ gives the sample size at which the isolation rate at resolution $r$ reaches a target $\varepsilon$,

$$n^*(\varepsilon, r) = \left(\frac{\log(1/\varepsilon)}{e^a}\right)^{1/b} r^{-d}. \tag{5}$$

The corpus enters through three numbers. The dimension $d$ sets how fast the required sample grows as the resolution becomes finer: each halving of $r$ multiplies $n^*$ by $2^d$. The constants $a$ and $b$ set the level and are fitted on the pilot.

### 3.3 Finite population correction

$G$ estimates the missing mass of the distribution that generated the corpus. When the estimand is the corpus itself, the $n$ sampled documents have distance zero to the sample, and each of the $N - n$ unsampled documents is isolated from the sample with probability close to $G$. The expected missing mass of the corpus is therefore

$$\hat{M}(n, r) = G(n, r)\,\big(1 - n/N\big). \tag{6}$$

The factor is the one survey sampling calls the finite population correction (Groves et al., 2009). There it scales the variance of a mean, and here it scales an expectation, because sampled units are covered by definition.

### 3.4 A diagnostic for corpora with more than one scale

The law assumes one dimension across the radii of interest, and a user must know before sampling whether it holds. The diagnostic uses the pilot alone. Within one pilot of 2,000 documents, we fit Equation 4 on subsamples of size up to 750 and compute $R^2$ on subsamples of size 1,000 to 2,000. A corpus passes when this pilot $R^2$ is at least 0.95. A second check needs no threshold: the ratio of coefficients from the two-regressor fit should agree with the TwoNN estimate of $d$. To validate the diagnostic we also compute the full out of sample $R^2$ on the whole corpus, from a fit on $n \le 1{,}000$ scored on $n$ up to 7,000, and call a corpus collapsing when it is at least 0.9. When the law fails, two probes locate the cause: the share of documents with a near duplicate (a nearest neighbour within cosine distance 0.02), and the spread of the TwoNN dimension across twenty $k$-means clusters.

## 4 Data

Table 1 lists the corpora with their results. Two synthetic families have a known answer. Gaussian manifolds place 20,000 points on a random $d$-dimensional subspace of $\mathbb{R}^{64}$ for $d \in \{2, 4, 8\}$, so the true intrinsic dimension is known. A Zipf topic mixture draws 20,000 documents from 200 topic centres with Zipf weights, isotropic Gaussian noise and normalisation to the unit sphere, so the topic of every document is known.

Ten real corpora come from four sources. Three forum corpora were assembled for earlier work on corpus summarization: AskDocs (medical questions from Reddit), Trump (posts about the 2020

election), and TIFU (stories told in the first person on Reddit), 10,000 documents each after cleaning and deduplication. They are embedded twice, with GTR-T5-base (Ni et al., 2022) at three random samples of the source (seeds 42 to 44) and with Qwen3-Embedding-8B (Zhang et al., 2025) at seed 42. Five corpora are keyword slices of a 2020 export of tweets about the United States Census: random, citizens, trump, covid and illegal, 7,610 to 30,136 tweets after URL and mention removal and deduplication of identical text. Two corpora are uniform random samples of 50,000 English posts each, drawn from collections of Truth Social and Bluesky posts between May and December 2024 and deduplicated on text, which also removes reposts that copy their original. The Truth Social collection holds the threads under posts about the 2024 candidates, and the Bluesky collection is a broad pool of political posts, so neither is a sample of its platform. All census, Truth Social and Bluesky corpora are embedded with Qwen3-Embedding-8B at 4,096 dimensions, truncated at 256 tokens and normalised. Cosine distance is used throughout. It is not a metric, and $\sqrt{2(1-\cos)}$, the Euclidean distance between normalised vectors, gives the same nearest neighbours.

The posts are public, and the analysis uses only their embeddings and corpus level statistics. No post or account is quoted or identified in the paper.

## 5 Results

### 5.1 The estimator tracks true topic coverage

The Zipf mixtures give every document a topic, so the true topic mass covered by a sample is known. Over four settings, three seeds and eleven sample sizes from 50 to 5,000, coverage $1-G$ at a radius of half the median pairwise distance of a pilot tracks the true coverage with Spearman $\rho = 0.968$ and a mean absolute error of 0.029 (Appendix A). Good-Turing on the true labels, which $G$ does not see, reaches $\rho = 0.989$. At smaller radii the agreement falls away. The radius matters, and the collapse law gives the dependence on it.

### 5.2 The isolation rate collapses on $nr^d$

Table 1 reports the fit of Equation 4 for 17 pairs of corpus and encoder. Ten pass the diagnostic: the three manifolds, AskDocs and TIFU under both encoders, the random census slice, Truth Social and Bluesky. On these ten, a fit on cells with $n \leq 1{,}000$ explains 92% or more of the variance of $\log(-\log G)$ on cells with $n$ up to 7,000. Fitting on $n \leq 300$ instead, a 23-fold extrapolation, keeps it above 0.92. Figure 1 shows the surfaces against $nr^d$.

The fit with $\log n$ and $\log r$ as separate regressors gives a second estimate of $d$ that does not use TwoNN. On the ten passing pairs the two estimates differ by at most 19%. On the seven failing pairs they differ by 27% to 43%, with the regression estimate always lower.

Two baselines predict $G$ at the largest $n$ from the cells with $n \leq 1{,}000$. A separate curve in $n$ at each radius, which needs no $d$, errs 1.5 to 3.8 times more than the collapse law on nine of the ten passing pairs, and less on the random census slice (0.025 against 0.040). The Chao and Jost extrapolation from singletons and doubletons (Chao and Jost, 2012), run within its recommended range of a doubling from $n = 1{,}000$ to 2,000, errs 2 to 9 times more than the collapse law on the passing pairs. Both baselines beat the collapse law on the Zipf mixture, which fails the diagnostic.

### 5.3 The sample size formula

For every pair, Equation 4 was fitted on cells with $n \leq 1{,}000$, and Equation 5 gave $n^*$ for targets $\varepsilon \in \{0.3, 0.2, 0.1\}$ at every radius where the mean surface crosses $\varepsilon$ beyond $n = 1{,}000$. The prediction was compared with the crossing interpolated on the log scale between grid points. On the ten passing pairs, 77 of 83 predictions fall within a factor of 1.5 of the empirical sample size. On the seven failing pairs, 10 of 23 do (Appendix B).

The error of $n^*$ comes from the shape of the law more than from the pilot. Resampling the pilot 50 times and refitting $d$, $a$ and $b$ each time gives intervals with a median width of 19% of $n^*$, and these cover the empirical crossing in only 21% of cases. We therefore report $n^*$ with the empirical band of a factor of 1.5, which held in 93% of cases on passing corpora.

A random sample can be compared with a sample chosen for coverage. Greedy farthest point selection, the core-set construction of Sener and Savarese (2018), minimises the largest distance from any document to the chosen set. To bring the missing mass at a radius below a target, a random sample needs 0.6 to 1.0 times as many documents as greedy selection on the forum and census corpora, and 0.12 times as many on Truth Social and Bluesky. Greedy selection spends its picks on outlying documents, which are few, while the missing

| Corpus | Encoder | $N$ | $d$ (sd) | $d$, ratio | $b$ | $R^2$ pilot | $R^2$ full | error, law | error, per radius |
|---|---|---|---|---|---|---|---|---|---|
| AskDocs | Qwen3 | 10,000 | 12.8 (0.39) | 12.2 | 0.54 | 0.98 | 0.99 | 0.015 | 0.057 |
| Truth Social | Qwen3 | 50,000 | 12.6 (0.15) | 14.4 | 0.43 | 0.98 | 0.98 | 0.025 | 0.053 |
| Census random | Qwen3 | 7,610 | 12.9 (0.39) | 12.4 | 0.27 | 0.98 | 0.97 | 0.040 | 0.025 |
| Manifold $d$=2 | none | 20,000 | 2.0 (0.05) | 2.4 | 0.80 | 0.97 | 0.96 | 0.027 | 0.065 |
| TIFU | Qwen3 | 10,000 | 13.6 (0.35) | 13.4 | 0.53 | 0.98 | 0.96 | 0.039 | 0.086 |
| Bluesky | Qwen3 | 50,000 | 16.0 (0.69) | 15.2 | 0.42 | 0.98 | 0.96 | 0.037 | 0.056 |
| TIFU | GTR | 10,000 | 18.3 (0.33) | 19.0 | 0.50 | 0.97 | 0.95 | 0.040 | 0.070 |
| Manifold $d$=8 | none | 20,000 | 8.0 (0.21) | 8.1 | 0.62 | 0.97 | 0.95 | 0.042 | 0.081 |
| AskDocs | GTR | 10,000 | 13.6 (0.36) | 13.6 | 0.51 | 0.98 | 0.95 | 0.042 | 0.087 |
| Manifold $d$=4 | none | 20,000 | 4.0 (0.16) | 4.3 | 0.72 | 0.96 | 0.92 | 0.036 | 0.091 |
| Census illegal | Qwen3 | 17,202 | 11.4 (0.33) | 7.5 | 0.26 | 0.93 | 0.85 | 0.026 | 0.053 |
| Census covid | Qwen3 | 17,837 | 10.7 (0.40) | 7.2 | 0.30 | 0.85 | 0.79 | 0.045 | 0.073 |
| Trump | Qwen3 | 10,000 | 9.0 (0.51) | 5.3 | 0.26 | 0.90 | 0.78 | 0.045 | 0.075 |
| Census trump | Qwen3 | 30,136 | 9.0 (0.84) | 6.3 | 0.26 | 0.87 | 0.78 | 0.043 | 0.056 |
| Census citizens | Qwen3 | 7,849 | 9.1 (0.24) | 6.6 | 0.27 | 0.76 | 0.73 | 0.043 | 0.058 |
| Trump | GTR | 10,000 | 9.0 (0.48) | 5.2 | 0.27 | 0.79 | 0.63 | 0.054 | 0.069 |
| Zipf, 200 topics | none | 20,000 | 11.3 (0.35) | 6.4 | 0.21 | 0.66 | 0.62 | 0.136 | 0.026 |

Table 1: The collapse law on 17 corpus and encoder pairs, sorted by the full out of sample $R^2$; the line separates the pairs that pass the diagnostic. $d$: TwoNN dimension, mean and sd over five pilots of 2,000 documents (and over three samples for the GTR forum corpora). $d$, ratio: the ratio of the $\log r$ and $\log n$ coefficients. $b$: exponent of Equation 4 fitted on cells with $n \leq 1{,}000$. $R^2$ pilot: fit on subsamples of one pilot with $n \leq 750$, scored on $n$ from 1,000 to 2,000. $R^2$ full: fit on $n \leq 1{,}000$, scored on $n$ up to 7,000. Errors: mean absolute error of the predicted $G$ at the largest $n$, for the collapse law and for a separate curve in $n$ at each radius.

| Corpus | near duplicates | local $d$ | CV of local $d$ |
|---|---|---|---|
| AskDocs (GTR) | 0.0% | 8.1 to 16.3 | 0.15 |
| TIFU (GTR) | 0.0% | 14.4 to 21.7 | 0.10 |
| Truth Social | 0.04% | 7.9 to 15.4 | 0.16 |
| Bluesky | 0.06% | 11.5 to 18.9 | 0.13 |
| Trump (GTR) | 3.4% | 2.0 to 16.2 | 0.52 |
| Zipf, 200 topics | 0.0% | 7.8 to 18.1 | 0.21 |

Table 2: Probes of the collapse failure. Near duplicates: share of a 5,000-document sample with a neighbour within cosine distance 0.02. Local $d$: range of the TwoNN dimension over twenty $k$-means clusters, and its coefficient of variation.

mass weighs documents equally.

### 5.4 The diagnostic

The diagnostic must work before the sample is drawn, so we compute it inside one pilot of 2,000 documents: fit Equation 4 on subsamples of size up to 750, and score $R^2$ on subsamples of size 1,000 to 2,000. The ten passing pairs reach a pilot $R^2$ of 0.96 or more, and the seven failing pairs reach at most 0.93 (Table 1). A pilot threshold of 0.95 therefore reproduces the full diagnostic on every pair. The threshold was chosen on these corpora, and the agreement of the two dimension estimates gives a second check that needs no threshold from them: on every failing pair the two differ by more than a quarter.

Section 3.2 says when the law should fail: when the effective dimension changes with the radius. On Trump the probes show such a change (Table 2). Its twenty $k$-means clusters differ most in local dimension: clusters of dimension near 2 hold slogans, and clusters near 16 hold discussion. At small radii the slogan clusters, where documents sit on a nearly flat set, dominate the count of isolated documents, and at large radii the discussion clusters do. Removing the 3.4% of documents with a near duplicate leaves the failure in place. The Zipf mixture changes dimension with radius for another reason: at small radii its documents look like noise around their centres, and at large radii they look like 200 points. The probes do not separate the failing census slices from the passing corpora, and we do not have a mechanism for them beyond their selection on a keyword that recurs in slogans and calls to action.

### 5.5 The finite population correction

Table 3 compares $G$ and $G\,(1 - n/N)$ with the true missing mass $M$ of the finite corpus on the eight forum and census corpora under Qwen3, at sampling fractions from 0.01 to 0.7 and three radii, 144 cells. Uncorrected, $G$ has a mean absolute error of 0.114 and a maximum of 0.49. Corrected, the mean error is 0.012 and the maximum 0.11, and at $n/N = 0.5$ the error is 0.004. On the three GTR forum corpora and the Zipf mixture the correction cuts the error from 0.068 to 0.015. The correction

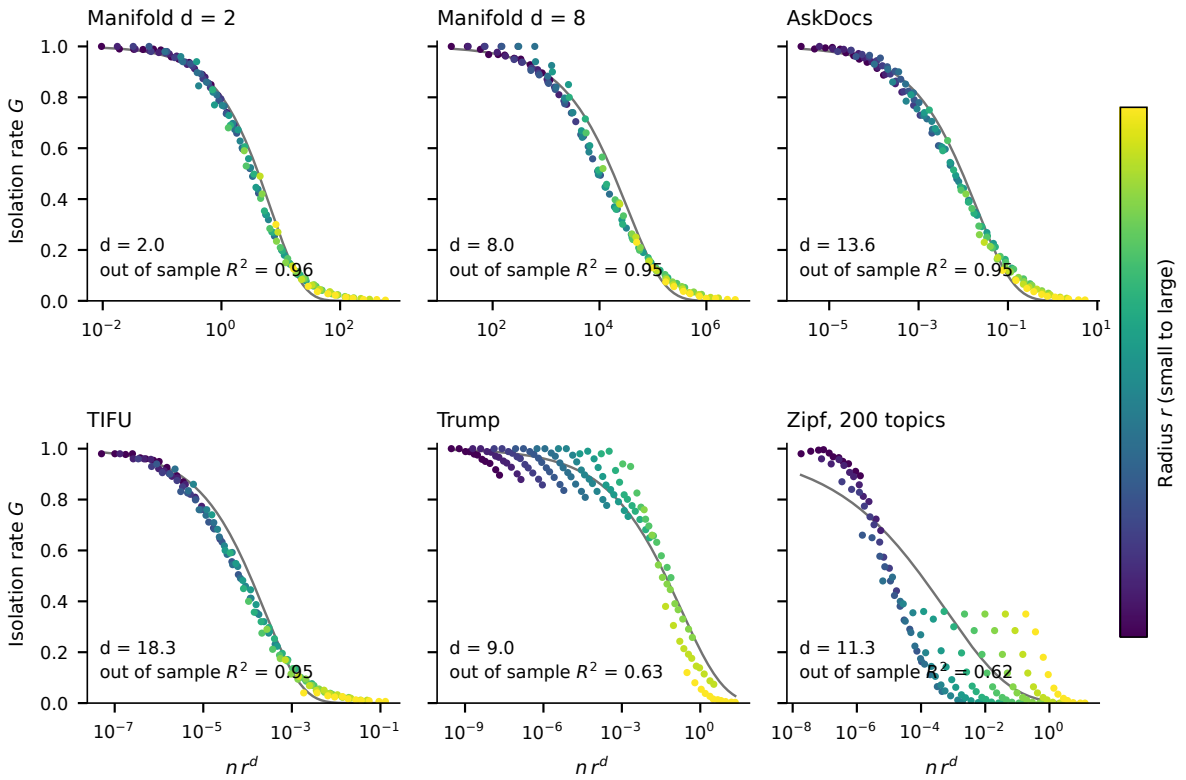



Figure 1: Isolation rate $G$ against $nr^d$ on a log scale, one panel per corpus (GTR encoder for the forum corpora), colour by radius $r$, one draw per cell. Points from different radii fall on one curve where the law holds (the manifolds, AskDocs, TIFU) and separate where it fails (Trump, the Zipf mixture). The grey line is the fit of Equation 4 on cells with $n \leq 1{,}000$; the annotation gives $d$ and the full out of sample $R^2$ of Table 1.

| $n/N$ | $M$ (true) | $G$ | $G\,(1 - n/N)$ | error |
|---|---|---|---|---|
| 0.01 | 0.762 | 0.768 | 0.760 | 0.032 |
| 0.05 | 0.590 | 0.623 | 0.592 | 0.014 |
| 0.10 | 0.497 | 0.547 | 0.492 | 0.010 |
| 0.25 | 0.347 | 0.459 | 0.344 | 0.006 |
| 0.50 | 0.198 | 0.396 | 0.198 | 0.004 |
| 0.70 | 0.110 | 0.366 | 0.110 | 0.003 |

Table 3: Missing mass of the finite corpus by sampling fraction, mean over eight Qwen3 corpora and three radii (the 25th, 50th and 75th percentiles of the pilot's nearest neighbour distances). The last column is the mean absolute error of the corrected estimate.

holds on the corpora that fail the collapse.

### 5.6 Choosing the resolution

The resolution $r$ is a cosine distance, and a user needs to know what content it groups. For each of the ten real corpora under Qwen3, we drew about 40 document pairs in each of ten distance bins from 0 to 1, 3,913 pairs in all, and asked Qwen3.8-27B whether the two documents are about the same specific content (the same claim, question, event or story), the same general topic, or unrelated (Appendix F). The share judged the same content falls with distance in every corpus. Where it crosses one half depends on the corpus, from $r = 0.05$ on TIFU to 0.32 on a census slice, and on Bluesky it stays below one half even for the nearest pairs (Table 4). Long first person stories differ in their details even when they are close in embedding space, while short posts that repeat a slogan are the same content at a larger distance. A cosine distance therefore does not carry one meaning across corpora, and a user who wants $r$ to mean "the same claim" should judge a few hundred pilot pairs binned by distance and take the distance where the share of same content crosses the chosen level.

### 5.7 What a target costs

With $r$ calibrated, Equation 5 says what a coverage target costs in each corpus. Table 4 gives the sample size at which half of the corpus lies within the same content distance of some sampled document. The answers span twelve orders of magnitude. On the census slices and the Trump forum corpus, whose posts repeat a few claims, a few dozen to about a thousand documents suffice, and

| Corpus | $d$ | $r_{\text{same}}$ | $n^*(0.5, r_{\text{same}})$ |
|---|---|---|---|
| TIFU | 13.6 | 0.05 | $4 \times 10^{14}$ |
| AskDocs | 12.8 | 0.15 | $1 \times 10^{9}$ |
| Truth Social | 12.6 | 0.16 | $1 \times 10^{7}$ |
| Census citizens | 9.1 | 0.23 | 480 (385) |
| Trump (forum) | 9.0 | 0.24 | 600 (622) |
| Census random | 12.9 | 0.28 | 1,100 (660) |
| Census covid | 10.7 | 0.29 | 350 (307) |
| Census illegal | 11.4 | 0.30 | 20 |
| Census trump | 9.0 | 0.32 | 50 |
| Bluesky | 16.0 | none | |

Table 4: The distance $r_{\text{same}}$ at which half of document pairs are judged the same specific content (Qwen3.8-27B, interpolated between bins of about 40 pairs), and the sample size the collapse law prescribes for an isolation rate of 0.5 at that distance, rounded. In parentheses, the empirical sample size where the grid reaches the target. On Bluesky 42% of pairs closer than 0.1 are judged the same content, so no distance qualifies.

where the grid reaches the target the empirical sample size is within a factor of two of the prediction. On AskDocs, TIFU and Truth Social the required sample exceeds the corpus by orders of magnitude: almost no document has another document in the corpus about the same specific content, and $G$ at the largest sample stays above 0.94. In those corpora no sample covers the specific content, and an analysis that needs every claim must read every document. The formula says which of the two situations a corpus is in before any annotation is bought.

## 6 Discussion

The procedure a user follows has five steps, and Section 5.7 shows its main use: deciding whether a coverage target is affordable at all. Choose the resolution $r$ that matches the question, by judging a few hundred pilot pairs binned by distance (Section 5.6). Embed a pilot of about 2,000 documents, estimate $d$ by TwoNN, and fit Equation 4 on subsamples of the pilot. If the pilot $R^2$ is at least 0.95 and the two dimension estimates agree, take $n^*$ from Equation 5 and allow a factor of 1.5 either way. If not, the corpus has more than one scale, and the probes of Section 5.4 say where. After the sample is drawn, report $G\,(1 - n/N)$ as the share of the corpus the sample leaves uncovered. No step needs labels or a GPU once the embeddings exist.

The survey analogy holds in one place and is instructive in two. The factor $1 - n/N$ is the same as the survey correction, applied here to an expectation. The dimension $d$ plays part of the role of the variance: it is the corpus property that sets how fast the required sample grows as the target becomes finer, and like the variance it is estimated from a pilot, here with two shape constants alongside it. The margin of error is the third place. A survey's margin bounds the error of a proportion, and a corpus sample's plain mean already does that job, to within 0.003 to 0.07 for cluster proportions (Appendix E). The missing mass bounds the mass of content the sample has no example of, which is the error that matters when the task is to list what is in the corpus. A report of a corpus sample can carry both numbers.

Coverage of a list of themes is a different quantity. When themes are a partition of the corpus with masses $m_1, \ldots, m_K$, the expected share of theme mass present in a sample of $n$ is $\sum_k m_k(1 - (1 - m_k)^n)$, set by the masses alone. With inventories of 10, 30 and 100 themes per corpus and with inventories of skewed mass, this expression matched the observed coverage of 1,200 samples from eight corpora to a mean error of 0.004, and the curves for a given inventory size nearly coincided across corpora (Appendix C). A theme inventory therefore measures coverage relative to its own granularity, and two corpora whose coverage at a fixed $r$ differs from 0.003 to 0.77 look the same through it (Appendix C). The stability of a free summary written from a sample is relative in the same way (Appendix D). Breadth at an absolute scale is visible only to a measure at that scale, which is what $G$ provides.

The diagnostic has a use beyond the formula. A corpus that fails it holds material at more than one scale, and a single sample size will oversample one part and undersample the other. The probe of local dimension across clusters locates the parts, and a sample stratified on them is the natural design. We have not tested it.

## Limitations

The resolution $r$ is a cosine distance in one encoder's space. The law held under both encoders on AskDocs and TIFU, but $d$ moved with the encoder (18.3 under GTR against 13.6 under Qwen3 on TIFU), so a user must fix the encoder before choosing $r$ and report the two together. Within one encoder the same $r$ also means different things in different corpora (Section 5.6), and the calibration rests on model judgments without a human check.

The required sample size grows as $r^{-d}$, so the

formula is useful only at resolutions where $n^*$ is a feasible number. At $d = 15$, halving $r$ multiplies $n^*$ by about 33,000. The formula plans samples at the resolution of the question and cannot make a fine resolution affordable.

The constants $a$ and $b$ are fitted on the pilot. We have an account of why $b$ is below 1 but no expression that predicts it.

The pilot threshold of 0.95 was chosen on the corpora of this paper, although the agreement of the two dimension estimates separates the same pairs without it. The formula's error band of a factor of 1.5 is empirical, and resampling the pilot does not reproduce it. The corpora are English social media and forum text, and the Truth Social and Bluesky collections are political pools, so their $d$ describes those collections.

The test unit is the document. Posts by the same account are correlated, and a design that samples accounts would need a cluster version of the estimator.

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## A Synthetic smoke test

The Zipf topic mixtures have 20,000 documents in 64 dimensions. Four settings vary the number of topics (50, 200 or 500), the Zipf exponent and the noise. Each setting was run at three seeds and eleven sample sizes from 50 to 5,000. Table 5 compares coverage $1 - G$ with the true topic mass coverage at radius $\alpha$ times the median pairwise distance of a pilot.

| $\alpha$ | Spearman $\rho$ | mean absolute error |
|---|---|---|
| 0.5 | 0.968 | 0.029 |
| 0.4 | 0.797 | 0.091 |
| 0.3 | 0.677 | 0.209 |
| 0.2 | 0.576 | 0.521 |
| 0.1 | 0.307 | 0.869 |

Table 5: Agreement of $1 - G$ with true topic coverage on the Zipf mixtures, 132 samples.

## B The sample size formula by corpus

Equation 4 was fitted on cells with $n \leq 1{,}000$, five draws per cell. Equation 5 then gave $n^*$ for $\varepsilon \in \{0.3, 0.2, 0.1\}$ at every radius where the mean surface crosses $\varepsilon$ at some $n > 1{,}000$, and the crossing was interpolated on the log scale between grid points. Table 6 reports the ratio of predicted to empirical sample size. The grid steps by a factor of about 1.5 between adjacent sample sizes, and the interpolated crossing varies across the five draws by a median of 6%.

## C Coverage of a theme inventory

Each of the eight forum and census corpora received inventories of themes by $k$-means on the full Qwen3 embeddings at $K = 10$, 30 and 100, and two inventories with skewed masses at $K = 30$: one from $k$-means++ seeding without refinement, one by merging the 100 clusters into 30 groups of Zipf sizes. The effective number of themes (the exponential of the entropy of the masses) ranges from 6.8 to 91. For each inventory and the 1,200 samples of Section D, Table 7 gives the range across corpora of the share of theme mass present. The expected coverage $\sum_k m_k(1 - (1 - m_k)^n)$ matches the observed value to a mean absolute error of 0.004 over all cells. For the $K = 30$ inventory, Gemma 4 (Gemma Team, 2026) was shown each sample and the list of named themes and asked which were present, with a supporting document for each. The mass of themes it reported rose with $n$ along the label coverage curve in every corpus, and the share of its reports that the labels confirm rose from 0.73 at $n = 25$ to 0.98 at $n = 400$.

| Corpus | Enc. | cases | median | $\leq 1.5$ | $\leq 2$ |
|---|---|---|---|---|---|
| AskDocs | Qwen3 | 5 | 1.04 | 5 | 5 |
| Truth Social | Qwen3 | 6 | 0.88 | 5 | 6 |
| Census random | Qwen3 | 3 | 0.81 | 2 | 2 |
| Manifold $d$=2 | none | 9 | 1.15 | 9 | 9 |
| TIFU | Qwen3 | 6 | 1.08 | 6 | 6 |
| Bluesky | Qwen3 | 5 | 0.74 | 3 | 4 |
| TIFU | GTR | 17 | 1.24 | 17 | 17 |
| Manifold $d$=8 | none | 7 | 1.12 | 7 | 7 |
| AskDocs | GTR | 16 | 0.96 | 14 | 16 |
| Manifold $d$=4 | none | 9 | 1.09 | 9 | 9 |
| Census illegal | Qwen3 | 3 | 1.24 | 2 | 3 |
| Census covid | Qwen3 | 3 | 1.22 | 3 | 3 |
| Trump | Qwen3 | 2 | 1.14 | 2 | 2 |
| Census trump | Qwen3 | 2 | 1.56 | 1 | 2 |
| Census citizens | Qwen3 | 2 | 2.06 | 0 | 1 |
| Trump | GTR | 6 | 1.86 | 2 | 3 |
| Zipf, 200 topics | none | 5 | 69 | 0 | 0 |

Table 6: Predicted over empirical sample size for a target isolation rate, in the order of Table 1. GTR rows pool three samples of the forum corpus.

| Inventory | $n$=25 | $n$=50 | $n$=100 |
|---|---|---|---|
| $k$-means, $K$=10 | 0.89 to 0.94 | 0.98 to 1.00 | 1.00 |
| $k$-means, $K$=30 | 0.58 to 0.64 | 0.82 to 0.85 | 0.94 to 0.98 |
| $k$-means, $K$=100 | 0.25 to 0.30 | 0.44 to 0.48 | 0.67 to 0.72 |
| seeding only, $K$=30 | 0.60 to 0.72 | 0.82 to 0.88 | 0.94 to 0.96 |
| Zipf merge, $K$=30 | 0.79 to 0.84 | 0.87 to 0.91 | 0.93 to 0.96 |

Table 7: Share of theme mass present in a sample, range across eight corpora.

## D Summary stability

For each of the eight forum and census corpora and each $n \in \{25, 50, 100, 200, 400\}$, thirty independent samples were summarized by Gemma 4 at temperature 0 in one paragraph of at most 150 words, and the summaries were embedded with Qwen3-Embedding-8B. Instability is the mean pairwise cosine distance among the thirty summaries. Within every corpus instability falls with $n$ and

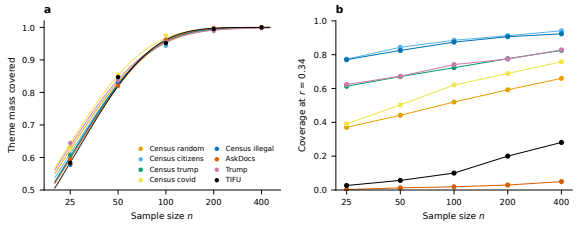



Figure 2: (a) Share of theme mass present in a sample against $n$ for the $K = 30$ inventory, eight corpora (points), with the expected coverage from each corpus's masses (lines). (b) Coverage $1 - G$ at the fixed radius $r = 0.34$ for the same samples.

tracks $G$ (Spearman $+0.40$ to $+1.00$, seven of eight at $+0.70$ or above). Across corpora at a fixed $n$, Spearman of instability with $G$ is $+0.14$, $+0.12$, $+0.12$, $+0.07$ and $-0.17$ at $n = 25$ to 400, with bootstrap bands that include zero. TIFU illustrates why: its $G$ is near 1 at the shared radius and its summaries are the most stable, because the model summarizes it as stories of mishap whichever thirty stories it is given.

## E A bound for functions that are stable at scale $r$

For a document function $f$ with range 1 that takes the same value on documents within $r$ of each other, the estimate of the corpus mean of $f$ formed by giving each corpus document the value of its nearest sampled document errs by at most $v(r) + M(r)$, where $v(r)$ is the share of covered documents whose nearest sampled document disagrees with them on $f$. On fifty cluster indicator functions over four corpora, six sample sizes and six radii, the bound held in 144 of 144 cells and was loose for proportions (median 22 times the realised error), where the plain sample mean already errs by 0.003 to 0.07.

## F Prompts

**Summary.** "Below are $\{n\}$ documents sampled from a larger collection. Write a summary of the collection in one paragraph of at most 150 words. Cover the main themes in proportion to how often they appear, name notable minority themes briefly, and do not mention individual document numbers."

**Theme naming.** "The $\{m\}$ documents below were grouped together because they are about the same theme. Give the theme a short name (at most 8 words) and a one-sentence definition that would let a reader decide whether a new document belongs to it. Answer as JSON with keys name and definition."

**Enumeration.** "Below are $\{n\}$ documents sampled from a larger collection, followed by a list of $\{K\}$ themes found in the full collection. Decide which themes are present among these $\{n\}$ documents. A theme is present only if you can point to one document that is primarily about it; in a small sample most themes will be absent, and similar-sounding themes are distinct, so pick the single best theme for each document. Answer with one line per present theme in the form theme_id: document_number."

**Same content.** "Two documents from the same collection are shown. Judge their relation. SAME: they are about the same specific content (the same claim, question, event or story), so a reader who has read one would learn little new from the other. TOPIC: they are about the same general topic but different specific content. UNRELATED: they are about different topics."